Dynamic Models of Firm Financing

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Roadmap

Many of the lectures you've seen are *method*-driven, while I will aim to be more *topic*-driven:

- Detailed examples of structural estimation in dynamic firm financing models
- A brief survey of related papers with similar model setups
- Speculations on future research directions (I find) particularly interesting

It all started with "Let's test how financially constrained firms are"...

1. Episode 1: Testing Euler equations
2. Episode 2: Fully blown dynamic investment models

Whited (1992): Debt, Liquidity Constraints, and Corporate Investment

Key idea: test Euler equations under financing constraints

- Write down a dynamic investment model with financing constraints
- Constraints introduce a **wedge** in the firm's Euler equation
- That wedge distorts the **intertemporal** allocation of investment

Empirical strategy (GMM on Compustat panel)

1. Estimate the investment Euler equation **with** vs. **without** a debt-constraint wedge
2. Split sample into constrained vs. unconstrained firms: Standard Euler fails more for the former (the wedge is larger for these firms)

- Q-theory fits poorly— perhaps because they ignore this wedge

Environment: value, dividends, and two constraints

Firm value = PDV of after-tax dividends (risk-neutral owners):

$$V_{i0} = \mathbb{E}_0 \sum_{t=0}^{\infty} \left(\prod_{j=0}^{t-1} \beta_{ij} \right) d_{it}.$$

- Capital law of motion: $K_{it} = I_{it} + (1-\delta)K_{i,t-1}$
  - s.t. a convex capital adjustment cost
- Sources-and-uses identity:
  $d_{it} = (1-\tau)\big(F(K_{i,t-1}, N_{it}) - w_t N_{it} - \Psi(I_{it}, K_{i,t-1}) - i_t B_{i,t-1}\big) + B_{it} - (1-\pi_t^e)B_{i,t-1} - P_{it} I_{it}$

Two financing constraints

1. Dividend nonnegativity: $d_{it} \geq 0$ (proxy for limited / costly outside equity); multiplier λ_{it}
2. Debt ceiling: $B_{it} \leq B_{it}^*$ (B^* taken as given by lenders); multiplier γ_{it}

Euler equations— equates marginal today to marginal tomorrow

1. Differentiate the Lagrangian w.r.t. K_{it} and B_{it}
2. Envelope condition of firm value w.r.t K_{it-1} and B_{it-1}
3. Expectation of 2. and substitute into 1.

Investment Euler — eq. (7):

$$\beta_{it}\,\mathbb{E}_t\left[\frac{1+\lambda_{i,t+1}}{1+\lambda_{it}}\left(F_K-\psi_K+(1-\delta)\left(\psi_I+\tfrac{p_{i,t+1}}{1-\tau}\right)\right)\right]=\psi_I+\tfrac{p_{it}}{1-\tau}$$

Do the same for debt — eq. (10) and substitute into eq. (7):

$$(1+\lambda_{it})-\beta_{it}\left(1+(1-\tau)i_t-\pi_t^e\right)\mathbb{E}_t(1+\lambda_{i,t+1})-\gamma_{it}=0.$$

Impose functional forms, can we now estimate these inter-temporal conditions?

Econometrics: proxy the debt wedge and split samples

Find proxies to how constrained firms are: (multipliers should move in the same direction)

$$\Lambda_{it} \equiv 1 - \frac{1+\lambda_{i,t+1}}{1+\lambda_{it}} = c_0 + c_1\,\mathrm{DAR}_{it} + c_2\,\mathrm{DAR}_{it}^2 + c_3\,\mathrm{COV}_{it} + c_4\,\mathrm{COV}_{it}^2,$$

- DAR: market debt/market assets (nearness to capacity)
- COV: interest/(interest + cash flow) (likelihood of distress)
- if firms are unconstrained, there should be no wedge, c's should all be 0

Two GMM exercises:

1. Are the c's jointly significant?
2. Performance of standard Euler ($\Lambda = 0$) and the augmented Euler by subgroups

Results: with vs. without bond ratings

Without bond ratings

	$\Lambda_{it} = 0$	$\Lambda_{it} = c(DAR_{it}, COV_{it})$		
α	0.619**	2.225**	1.156**	1.567**
(Adjustment cost parameter)	(0.094)	(0.674)	(0.532)	(0.813)
μ	0.014	1.717**	1.082**	1.049**
(Mark-up)	(0.011)	(0.923)	(0.196)	(0.067)
η	0.420	0.831	0.988**	0.515
(Returns-to-scale parameter)	(0.557)	(1.335)	(0.161)	(0.319)
v^2	–	2.620	1.875	0.846
("Normal" rate of investment)		(3.892)	(1.454)	(0.786)
c_0 (Constant)	–	0.0061	0.0020	0.0073*
		(0.0044)	(0.117)	(0.0044)
c_1 (DAR_{it})	–	0.252**	0.442**	–
		(0.097)	(0.172)	
c_2 (DAR^2_{it})	–	0.0022*	0.013	–
		(0.0013)	(0.094)	
c_3 (COV_{it})	–	0.039**	–	0.055**
		(0.012)		(0.014)
c_4 (COV^2_{it})	–	0.0136*	–	0.069
		(0.0079)		(0.042)
χ^2 (Overidentifying restrictions)	38.124	6.033	7.394	8.970
Degrees of freedom	9	3	5	5
p-Value	1.66×10^{-5}	0.110	0.193	0.110
χ^2 (Exclusion restrictions on Λ_{it})	–	32.091	30.730	29.154
Degrees of freedom	–	6	4	4
p-Value	–	1.57×10^{-5}	3.48×10^{-6}	7.27×10^{-6}

With bond ratings

	$\Lambda_{it} = 0$	$\Lambda_{it} = c(DAR_{it}, COV_{it})$		
α	1.734**	1.998**	0.746**	1.238**
(Adjustment cost parameter)	(0.663)	(0.461)	(0.327)	(0.445)
μ	1.450**	1.100**	1.279**	1.191**
(Mark-up)	(0.137)	(0.084)	(0.314)	(0.107)
η	0.598	0.982	1.224	0.669
(Returns-to-scale parameter)	(0.612)	(0.855)	(0.899)	(0.401)
v^2	–	1.003	0.882*	0.900
("Normal" rate of investment)		(4.948)	(0.513)	(0.871)
c_0 (Constant)	–	0.0007	0.0005	0.0015
		(0.0012)	(0.0021)	(0.0024)
c_1 (DAR_{it})	–	0.124*	0.113	–
		(0.072)	(0.089)	
c_2 (DAR^2_{it})	–	0.0008	0.0081*	–
		(0.0074)	(0.0050)	
c_3 (COV_{it})	–	0.011*	–	0.061*
		(0.007)		(0.038)
c_4 (COV^2_{it})	–	0.0064	–	0.0084
		(0.0043)		(0.012)
χ^2 (Overidentifying restrictions)	21.004	4.577	6.080	10.375
Degrees of freedom	9	3	5	5
p-Value	0.013	0.206	0.298	0.065
χ^2 (Exclusion restrictions on Λ_{it})	–	16.427	14.924	10.629
Degrees of freedom	–	6	4	4
p-Value	–	0.012	0.0049	0.031

- Standard Euler is rejected for *both* groups, but **more** for firms without bond ratings

Results: financially healthy vs. unhealthy firms

Healthy firms

	Low-Debt Firms			Low-Coverage Firms		
α	0.956**	1.051**	1.312**	1.742**	1.244*	1.646**
(Adjustment cost parameter)	(0.222)	(0.302)	(0.632)	0.573	(0.691)	(0.872)
μ	1.033**	1.218**	1.322**	1.567**	1.437**	1.438**
(Mark-up)	(0.078)	(0.149)	(0.181)	(0.439)	(0.694)	(0.407)
η	0.998	0.839	0.814	0.643	0.140	0.632
(Returns-to-scale parameter)	(0.712)	(0.710)	(1.132)	(0.520)	(0.710)	(0.694)
v^2	1.172	1.716	0.870	1.490	0.552	1.005
("Normal" rate of investment)	(0.691)	(1.331)	(0.629)	(1.539)	(0.824)	(0.621)
c_0 (Constant)	0.0009	0.0009	0.0069	0.0099	0.0012	0.0013
	(0.0028)	(0.0016)	(0.0043)	(0.014)	(0.0008)	(0.0218)
c_1 (DAR_{it})	0.100*	0.094	–	0.381	0.102	–
	(0.053)	(0.168)		(0.200)	(0.071)	
c_2 (DAR^2_{it})	0.0012	0.0073*	–	0.0002	0.0065	–
	(0.0058)	(0.0044)		(0.0054)	(0.0046)	
c_3 (COV_{it})	0.023	–	0.096	0.016*	–	0.055
	(0.038)		(0.088)	(0.0087)		(0.064)
c_4 (COV^2_{it})	0.0047	–	0.0092	0.0029	–	0.0014
	(0.0041)		(0.013)	(0.0097)		(0.0076)
χ^2 (Overidentifying restrictions)	6.959	6.037	8.719	7.503	7.560	8.137
Degrees of freedom	3	5	5	3	5	5
p-Value	0.073	0.303	0.121	0.057	0.182	0.149
χ^2 (Exclusion restrictions on Λ_{it})	2.458	3.380	0.698	0.949	0.892	0.315
Degrees of freedom	6	4	4	6	4	4
p-Value	0.873	0.496	0.952	0.987	0.926	0.989

Unhealthy firms

	High-Debt Firms			High-Coverage Firms		
α	1.056**	1.261**	0.526*	1.123**	1.326**	1.017**
(Adjustment cost parameter)	(0.511)	(0.308)	(0.292)	(0.490)	(0.382)	(0.364)
μ	1.309**	1.108**	0.977**	0.998**	1.113**	1.403**
(Mark-up)	(0.245)	(0.269)	(0.150)	(0.155)	(0.304)	(0.024)
η	1.425*	0.905	0.900	0.805	0.862	0.461
(Returns-to-scale parameter)	(0.794)	(0.533)	(1.112)	(0.522)	(0.889)	(0.411)
v^2	2.764	1.753	0.832	2.622	0.873	0.662
("Normal" rate of investment)	(1.926)	(0.974)	(0.536)	(1.916)	(1.540)	(0.724)
c_0 (Constant)	0.0089	0.0009	0.0031	0.0091*	0.0052	0.0045*
	(0.0062)	(0.0043)	(0.0029)	(0.0051)	(0.0031)	(0.0026)
c_1 (DAR_{it})	0.241**	0.426**	–	0.364**	0.343**	–
	(0.108)	(0.096)		(0.147)	(0.167)	
c_2 (DAR^2_{it})	0.0025	0.025	–	0.016**	0.103	–
	(0.049)	(0.471)		(0.0052)	(0.068)	
c_3 (COV_{it})	0.094**	–	0.021**	0.055*	–	0.060
	(0.049)		(0.0066)	(0.031)		(0.054)
c_4 (COV^2_{it})	0.0004	–	0.081	0.0005	–	0.055
	(0.0003)		(0.927)	(0.014)		(0.029)
χ^2 (Overidentifying restrictions)	6.958	6.645	9.433	8.997	5.952	8.758
Degrees of freedom	3	5	5	3	5	5
p-Value	0.073	0.248	0.019	0.029	0.311	0.119
χ^2 (Exclusion restrictions on Λ_{it})	14.270	14.583	11.795	20.654	23.699	20.893
Degrees of freedom	6	4	4	6	4	4
p-Value	0.027	0.0056	0.093	0.0021	9.18×10^{-5}	3.33×10^{-4}

- In the augmented Euler, DAR/COV matter for unhealthy firms, not for healthy ones

DeAngelo, DeAngelo, and Whited (2011)

Shortcomings of Episode 1

- The Euler approach is mainly a way to *describe* features of the data
- It does not fully trace out firm behavior given the economic environment

Capital structure dynamics and transitory debt

- Builds on Hennessy and Whited (2005, 2007): fully dynamic investment *and* financing
- Firm chooses investment and next-period net debt to balance **today's funding need** against the option to borrow again after future shocks

Environment: production and capital

Technology

$$\pi(k,z) = z\,k^{\theta}, \qquad 0 < \theta < 1, \qquad \ln z' = \rho \ln z + \nu'.$$

$$I = k' - (1-\delta)k, \qquad A(k,k') = \gamma k\,\mathbf{1}_{\{I\neq 0\}} + \frac{\alpha}{2}\left(\frac{I}{k}\right)^2 k.$$

- z and k: productivity shock (AR(1) in logs) and capital
- Investment: $I = k' - (1-\delta)k$
- Adjustment costs: **fixed** (γ) and **convex** (α)
- Then add net debt p to the state: (k, p, z) $\quad$ ($p > 0$ debt; $p < 0$ cash; never both)

Environment: three costly sources of funds

1. **Debt capacity** — riskless perpetual debt, finite limit:

$$p \leq \bar{p}$$

2. **Agency / carrying cost of cash** ($p < 0$):

$$s(p) = s\,|p|\,\mathbf{1}_{\{p<0\}}$$

3. **Equity issuance cost** — linear–quadratic; $e < 0$ means issue:

$$f(e) = \mathbf{1}_{\{e<0\}}\left(\lambda_1 e - \tfrac{1}{2}\lambda_2 e^2\right)$$

Environment: value function

$$V(k,p,z) = \max_{k',p'} \left\{ e + f(e) + \frac{1}{1+r} \int V(k',p',z')\, g(dz'|z) \right\}$$

subject to sources-and-uses of funds condition (corporate tax τ_c):

$$e = (1-\tau_c)\pi(k,z) + p' - p\big(1 + r(1-\tau_c)\big) + \delta k\, \tau_c - I - A(k,k') - s(p).$$

We can directly solve the value/policy function

- **Value function** V: maximized equity value — the PV of net distributions under optimal behavior
- **Policy function** $\{k', p'\}$: optimal investment and next-period net debt in each state
- ...depends on current state (k,p,z), and the environment

FOC: marginal value of one dollar of debt today

What is the marginal effect on firm value of an extra dollar of net debt p?

Envelope ($m =$ multiplier on $p \leq \bar{p}$):

$$\underbrace{-V_p}_{\substack{\text{marginal value}\\ \text{of debt capacity today}}} = \underbrace{\left(1 + (1-\tau_c)r - s\,\mathbf{1}_{\{p<0\}}\right)}_{\substack{\text{gross return on cash}\\ \text{(or borrowing cost)}}} \left(\underbrace{-EV_{p'}}_{\substack{\text{expected marginal value}\\ \text{of debt capacity tomorrow}}} + \underbrace{\lambda}_{\substack{\text{shadow value of relaxing}\\ \text{the collateral constraint}}} \right).$$

- So $-V_p$ is the cost (benefit) of carrying \$1 more debt (debt capacity) on the books
- What is special? it is fully forward looking

Spare debt capacity

Do firms always exhaust $\bar{p}$?: think of a firm hit by a positive TFP shock...

- No — positive TFP shock today means high return from investment tomorrow
- The financing cost is likely high tomorrow
- Increase incentive to preserve debt capacity today

Will the firm hold too much cash—that the tradeoff above is not relevant?

- No — holding cash entails a "carrying cost" from: agency friction in this model
- corporate tax adds to the cost of carrying cash

- Link to Episode 1: the financing wedge is *state-dependent*—the collateral constraint does not have to be binding today

Application: how firms adjust leverage

Standard tradeoff theory

- Firms have a target leverage ratio
- When leverage differs from target, they rebalance *toward* it as fast as issuance costs allow

Empirical challenge

- SOA is "suspiciously slow" (Fama–French 2002; Flannery–Rangan; Lemmon–Roberts–Zender;...)
- $\Rightarrow$ doubt on tradeoff theories (or issuance costs would have to be extremely large)

What is "target leverage"

Modigliani–Miller tradeoff: tax shield vs. bankruptcy costs (+ information / agency costs in richer versions).

How to think about target leverage

If we view DDW statically: is there an interior target leverage?

No. With linear tax benefits and no bankruptcy cost, the static problem delivers a **corner**—not an interior target.

So can we still define a "target" in DDW?

- Yes, but it is an 'addon", not innate to the model (what *is* innate then?)

But still doable... think about the following thought experiment:

- The firm optimizes under uncertainty
- By chance, it faces an arbitrarily long sequence of *neutral* investment shocks
- The leverage ratio to which it converges $\equiv$ **target**

Estimation sketch: SMM

Why estimate?

Quantitative predictions are credible only if parameters are disciplined by data

Procedure Solve the model numerically $\rightarrow$ simulate panels under the optimal policy $\rightarrow$ match moments to data

Estimated parameters:

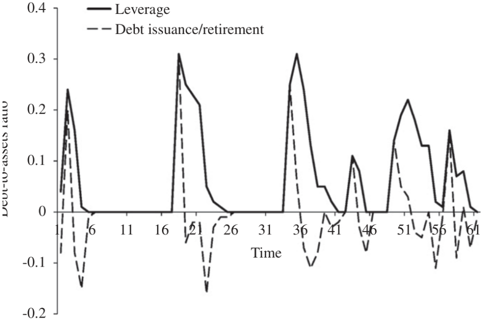
- θ, ρ, σ_ν: production side parameters
- α, γ: capital adjustment costs
- s: agency cost of cash
- λ_1, λ_2: equity issuance costs
- $\bar{p}/k_{ss}$: debt capacity

Moments: real (investment, CF volatility, serial correlation,...) and financial (leverage, cash, issuance frequencies/ sizes,...). Do *not* target how fast leverage adjusts to target.

A special case: no tax benefit (DDW Fig. 1)

Shut off the tax advantage of debt $\Rightarrow$ **target leverage** $= 0$

- Leverage rises with funding needs, then falls with a lag
- Return to target can take many periods if shocks keep arriving
- Firms with positive leverage can deviate even more from the "target" before returning



DDW (2011), Fig. 1: realized leverage with target $= 0$

Back to the speed of adjustment

Estimation yields low speed of adjustment that lands in the empirical range:

- $\approx 1/3$ to $1/12$ per year, which has been considered "too slow" in a tradeoff model

Mechanisms:

- Investment and capital structure are jointly determined; treat investment opportunities as a first-order driver of savings/borrowing (I will come back to this point later)
- SOA is dampened by intentional temporary movements *away* from target to fund investment.
- Conventional SOA understates the strength of firms' true rebalancing incentives when such incentives are present

Using a structural estimation to rationalize an interesting statistical relationship

Slow SOA $\neq$ weak tradeoff motives

As σ_ν rises: lower targets, more cash, large transitory debt

		Standard deviation of investment shocks:							
		Low			Moderate				High
1.	Average investment (I/k)	0.158	0.160	0.163	0.167	0.169	0.171	0.172	0.171
2.	Standard deviation of investment (I/k)	0.129	0.145	0.166	0.187	0.197	0.211	0.214	0.213
3.	Frequency of investment	0.852	0.833	0.791	0.775	0.763	0.754	0.767	0.772
4.	Average debt-to-assets ratio (d/k)	0.722	0.508	0.336	0.203	0.133	0.104	0.091	0.091
5.	Standard deviation of leverage (d/k)	0.110	0.089	0.101	0.096	0.087	0.077	0.072	0.069
6.	Average net debt $((d-c)/k)$	0.722	0.508	0.333	0.190	0.081	0.011	−0.010	−0.007
7.	Standard deviation of net debt	0.110	0.089	0.127	0.145	0.227	0.272	0.261	0.248
8.	Average cash balances to assets (c/k)	0.000	0.000	0.002	0.010	0.046	0.085	0.091	0.088
9.	Standard deviation of (c/k)	0.000	0.000	0.083	0.153	0.303	0.306	0.269	0.236
10.	Frequency of positive debt outstanding	1.000	1.000	0.965	0.908	0.779	0.672	0.639	0.627
11.	Average of positive leverage values	0.722	0.508	0.348	0.223	0.171	0.155	0.142	0.146
12.	Average of positive cash balance values	0.000	0.000	0.091	0.165	0.283	0.334	0.322	0.302
13.	Debt issuance frequency	0.468	0.463	0.451	0.438	0.393	0.341	0.330	0.328
14.	Debt repayment frequency	0.521	0.534	0.528	0.499	0.436	0.390	0.367	0.356
15.	Average debt issuance/assets	0.088	0.092	0.106	0.114	0.106	0.111	0.106	0.104
16.	Average debt repayment/assets	−0.069	−0.070	−0.076	−0.075	−0.069	−0.067	−0.064	−0.064
17.	Equity issuance frequency	0.255	0.257	0.261	0.255	0.245	0.237	0.239	0.236
18.	Average equity issuance/assets	0.028	0.024	0.022	0.018	0.019	0.018	0.018	0.017
Average fraction of investment funded from:									
19.	Current cash flow	0.840	0.847	0.832	0.823	0.816	0.802	0.807	0.810
20.	Cash balances	0.000	0.000	0.004	0.017	0.047	0.076	0.083	0.086
21.	Debt issuance	0.143	0.135	0.148	0.148	0.125	0.112	0.098	0.093
22.	Equity issuance	0.017	0.018	0.016	0.012	0.012	0.009	0.011	0.011
23.	Target	0.748	0.503	0.304	0.032	0.000	0.000	0.000	0.000
24.	Incidence of leverage above target	0.398	0.645	0.708	0.890	0.779	0.675	0.642	0.631
25.	Average deviation from target	−0.027	0.005	0.032	0.170	0.133	0.104	0.091	0.092
26.	Average transitory debt	0.065	0.057	0.092	0.195	0.170	0.155	0.142	0.146

High ρ, profitability, or lumpiness: lower targets, more cash

		Shock serial correlation		Marginal profitability		Optimal investment	
		Low	High	Low	High	Smooth	Lumpy
1.	Average investment (I/k)	0.151	0.178	0.161	0.164	0.158	0.251
2.	Standard deviation of investment (I/k)	0.051	0.244	0.150	0.171	0.129	0.742
3.	Frequency of investment	0.998	0.816	0.828	0.772	0.872	0.426
4.	Average debt-to-assets ratio (d/k)	0.889	0.085	0.711	0.094	0.400	0.088
5.	Standard deviation of leverage (d/k)	0.038	0.084	0.192	0.083	0.087	0.134
6.	Average net debt ($(d-c)/k$)	0.889	-0.453	0.711	-0.004	0.399	-0.184
7.	Standard deviation of net debt	0.038	1.413	0.192	0.258	0.089	0.669
8.	Average cash balances to assets (c/k)	0.000	0.530	0.000	0.090	0.000	0.207
9.	Standard deviation of (c/k)	0.000	1.894	0.000	0.252	0.023	0.811
10.	Frequency of positive debt outstanding	1.000	0.527	1.000	0.644	0.998	0.456
11.	Average of positive leverage values	0.889	0.161	0.711	0.146	0.400	0.194
12.	Average of positive cash balance values	0.000	1.256	0.000	0.318	0.083	0.528
13.	Debt issuance frequency	0.024	0.263	0.420	0.331	0.489	0.217
14.	Debt repayment frequency	0.017	0.272	0.437	0.385	0.503	0.390
15.	Average debt issuance/assets	0.027	0.100	0.030	0.101	0.060	0.564
16.	Average debt repayment/assets	-0.034	-0.059	-0.028	-0.065	-0.052	-0.138
17.	Equity issuance frequency	0.165	0.293	0.061	0.253	0.248	0.248
18.	Average equity issuance/assets	0.016	0.039	0.020	0.015	0.020	0.032
Average fraction of investment funded from:							
19.	Current cash flow	0.987	0.809	0.975	0.786	0.899	0.671
20.	Cash balances	0.000	0.092	0.000	0.090	0.000	0.126
21.	Debt issuance	0.002	0.067	0.022	0.114	0.089	0.198
22.	Equity issuance	0.011	0.028	0.003	0.009	0.011	0.005
23.	Target	0.904	0.000	0.702	0.000	0.371	0.000
24.	Incidence of leverage above target	0.037	0.532	0.501	0.643	0.712	0.593
25.	Average deviation from target	-0.015	0.085	0.009	0.094	0.035	0.129
26.	Average transitory debt	0.012	0.160	0.166	0.146	0.084	0.217

Extension 1: Defaultable debt

Relative to DDW's risk-free debt with an exogenous ceiling:

- Debt is **defaultable** $\Rightarrow$ credit spread; cost of debt $\neq$ risk-free rate
- Default when equity value hits zero (limited liability); recovery $\mathcal{R} <$ face value

Debt pricing:

$$q(k', b', z)\, b' \;=\; \frac{1}{1+r}\, \mathbb{E}_{z'|z}\Big[\underbrace{\mathbf{1}\{\text{no default}\}\; b'}_{\text{repaid in full}} + \underbrace{\mathbf{1}\{\text{default}\}\; \mathcal{R}(k', z')}_{\text{recovery value}}\Big]$$

$\Rightarrow$ higher leverage raises default risk, lowers q: the credit spread is *endogenous*

It behaves a lot of times like a collateral constraint, why?

- Financing wedge is highly nonlinear, firms choose not to borrow beyond a given point

Extension 1: Defaultable debt

Two implementations

- **Hennessy and Whited (2007)**: *"How Costly Is External Financing?"*
- **Begenau and Salomao (2019)**: *"Firm Financing over the Business Cycle"*

What they share:

- Both frictions: defaultable debt *and* costly equity issuance
- Both contrast **small vs. large firms** as the key heterogeneity

Where they differ:

- **HW**: *idiosyncratic* TFP; focus on cross-firm differences, targeting cross-sectional mmt
  - esp, across established measures of financial constraint proxies
- **BS**: + *aggregate* TFP; focus on financing cross business cycles and match cyclical mmt

Extension 2: Equity misvaluation

Relative to DDW's correctly priced equity, in reality:

- Market equity value can deviate from fundamentals (**misvaluation**)
- Managers exhibit market-timing incentives

Introduce ψ:

How firm value deviates from fundamentals? $\psi > 1 (< 1)$ is over-(under-) valuation

$$\ln \psi' = \mu_\psi + \rho_{z\psi} \ln z + \rho_\psi \ln \psi + \varepsilon_\psi, \qquad \varepsilon_\psi \sim \text{i.i.d. } N(0, \sigma_\psi^2)$$

$$V(z, \phi, K, C) = \max_{K', C', E} (1 - \tau_d) D + \beta \mathbb{E}\left[\frac{f'}{f} V(z', \phi', K', C')\right]$$

with dilution factor $$\frac{f'}{f} = \frac{V - (1 - \tau_d) D}{V - (1 - \tau_d) D + E/\psi}$$

Extension 2: Equity misvaluation

Why does mispricing move real decisions?

- Overvaluation makes equity cheap: issue, then invest or save; undervaluation favors repurchases, which can crowd out investment
- Both represent market-timing opportunities, and increase firm valuation
- Repurchases are no longer “equivalent” to dividends

Two implementations

- **Warusawitharana and Whited (2016)**: —market timing contributes quite strongly to firm value, modestly to investment
- **Choi, Tian, Wu, and Kargar (2025)**: microfound misvaluation via *investor demand* (demand approach, portfolio-holdings data); similar results as in WW, more heterogeneity across firms $\Rightarrow$ *capital misallocation*

Extension 3: Agency conflicts

Firm policies are controlled by the **manager**, who has their own private incentives

- **Nikolov and Whited (2014)**: profit share α, equity stake β, diversion s

$$u \;=\; (\alpha + s)\,\varepsilon k^{\theta} \;+\; s\,c(1+r) \;+\; \beta\,d$$

- **Glover and Levine (2015)**: stock θ_s, options θ_o, fixed salary F; CRRA $U(c) = \frac{c^{1-\gamma}}{1-\gamma}$

$$C \;=\; (1+r_f)\,\theta_s\,d \;+\; F \;+\; \theta_s\,V' \;+\; \theta_o \max\big(V' - S,\, 0\big)$$

Manager value (NW Bellman; GL: max $\mathbb{E}[U(C)]$):

$$U(k, c, \varepsilon) \;=\; \max_{k',\, c'} \left\{ u \;+\; \frac{1}{1+r}\, \mathbb{E}\big[U(k', c', \varepsilon')\big] \right\}$$

Firm value is *derived* from the manager's policy:

$$V \;=\; d \;+\; \frac{1}{1+r}\, \mathbb{E}\big[V'\big]$$

Extension 3: Agency conflicts

Agency friction can come from observed compensation contracts

They can also come from unobserved factors... which need to be estimated

Two implementations

- **Terry (2023)**: managers derive private benefit from R&D, but face a cost from missing short-term earnings target (θ_π clawback if $\pi < \pi^f$)
- **Bordalo, Gennaioli, Shleifer, and Terry (2026)**: diagnostic expectations ($\mathbb{E}^\theta$)

Unlike the previous case, these incentives are not directly observable...

How to identify these incentives?—Introduce additional sources of data:

- **Terry (2023)**: analyst *earnings-forecast*: moments on meeting /bunching /R&D–profit comovement
- **BGST (2026)**: *manager own forecast* : their forecast errors and comovement with firms' investment or debt

Extension 4: Banks (Van den Heuvel 2002)

Same dynamic investment–financing problem—now for a **bank**

- Paper is about monetary transmission; we skip that and focus on the *setup*, especially the **precautionary** motive
- "Capital stock" $=$ loan book L; "investment" $=$ new lending N
- Funded with short debt/deposits B and equity E $\quad (L = B + E)$
- Capacity constraint: previously collateral / debt ceiling; here a regulatory equity ratio

$$E \geq \gamma L$$

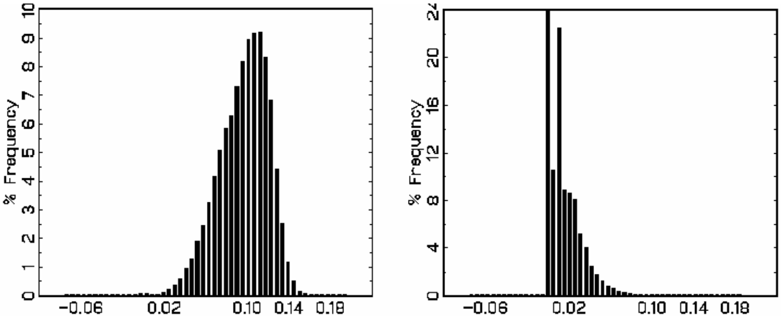
Precautionary buffer (same logic as spare debt capacity)

Banks hold *excess capital* even when the constraint is slack today— anticipating that it may bind after future shocks

Extension 4: DDW firm $\leftrightarrow$ Van den Heuvel bank

DDW firm	**Van den Heuvel bank**
Profit from capital: $\pi(k, z) = z\, k^{\theta}$	Profit from loans: $\pi \approx \rho L + rB -$ charge-off (shock)
Accumulation of k: $k' = (1-\delta)k + I$	Accumulation of L: $L' = (1-\delta')(1-\omega')L + (1-\delta')(1-b')N$
Financing of k: debt p, cash, equity e (if any)	Financing of L: short debt $B = L - E$; equity capital
Debt capacity: $p \leq \bar{p}$	Equity ratio: $E \geq \gamma L$
Equity issuance is costly $f(e)$	Equity issuance is shut down: $D \geq 0$
Spare debt capacity	Excess capital buffer

Extension 4: Excess capital (precautionary buffer)



Van den Heuvel Fig. 1: begin-of-period (left) vs. post-decision (right)

- **Begin-of-period:** mass well above zero
- **Post-decision:** closer to constraint, still > 0

Extension 5: Precautionary savings—two economically distinct sources

Recall: what gives rise to precautionary motives in DDW?

- Investment under financing constraints (uncertain future funding needs)
- Precautionary savings enable investment, which generates resources—not from risk aversion
- Firm is risk-neutral

Another type: Kaboski–Townsend (2011)

- Households face **incomplete insurance** and **borrowing constraints**
- Precautionary savings come from the desire to smooth consumption: the "curvature" is concave $u(c)$ (prudence)

"First order" investment effect vs. "second order" consumption smoothing effect

- This is *not* about importance. It is about order of magnitude in a small-risk /Taylor sense

Kaboski–Townsend (2011): Setup

Agents: rural households choose consumption, saving/borrowing, and (lumpy) investment

$$Y_{t+1} = U_{t+1}\big(P_t\, G\, N_{t+1} + R\, D_t\, I_t^*\big), \qquad L_{t+1} = Y_{t+1} + S_t(1+r)$$

- P: permanent income; U, N: transitory / permanent shocks
- S: liquid saving ($S < 0 =$ borrowing);
- L: liquid wealth
- $D \in \{0,1\}$: take project of size $I^* = i^* P$; fixed return $R > r$

Household problem

$$\max_{\{c_t, S_t, D_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t\, u(c_t) \quad \text{s.t.} \quad c_t + S_t + D_t I_t^* \leq L_t, \quad S_t \geq \underline{s}\, P_t$$

Kaboski–Townsend: precautionary savings and empirics

Consumption smoothing: concave u + incomplete insurance+ borrowing limit $S_t \geq \underline{s}\, P_t$.

Savings also enable lumpy investment with $R > r$—similar to DDW, but how important?

- how often it jumps the threshold
- how profitable the investments are
- how "correlated" the shocks are
- in conclusion, authors note that less transient projects would raise "anticipatory savings"

Step 1: Estimate the model

Pre-program Townsend Thai panel; SMM on consumption, income, investment, default, and credit moments $\leftrightarrow$ preferences, shocks, and $\underline{s}$.

Step 2: Million Baht as a constraint loosening

Same transfer per village $\Rightarrow$ exogenous credit per household. Surprise rise in $\underline{s}$;
Compare model responses to the reduced form, focusing on consumption

A digression: combining structural and reduced-form strategies

KT uses a field experiment to validate a structural model (buffer-stock response to a credit-limit relaxation).

Same idea in firm finance—**Li–Whited–Wu (2016)**: antirecharacterization laws as a quasi-natural experiment.

- Strengthen SPV secured-borrowing rights in seven states $\Rightarrow$ looser collateral constraints
- Estimate a DDW-like model on pre/post $\times$ treated/control
- Recover how much the collateral constraint moved
- Check simulated responses against the reduced form

Two ways to combine

1. **Validation / mapping** (KT, LWW)
2. **Direct targets:** put identified treatment effects into the SMM/GMM objective

Do we have a model with both? Boar–Gorea–Midrigan (2026)

So far the two motives lived in separate models:

- **DDW**: a real firm (k, debt capacity), but owners are risk-neutral
- **KT**: risk-averse households, but investment side is relatively "light"

The private business owner is *both*

Undiversified entrepreneur: the same agent chooses the firm's scale *and* bears the profit risk in their own consumption.

Motivating facts (Orbis Spain + Survey of Household Finances):

- Profit shares swing hugely: mean 0.13, yet 5% of firms lose 20% of output
- Ownership is concentrated: 93% of entrepreneurs own one business; average ownership share 83%

Boar–Gorea–Midrigan: Setup

States (m, z): cash-on-hand, productivity. Each period pick an **occupation**: worker or entrepreneur.

$$V(m,z) = \max\{V^w,\ V^e\}, \qquad V^e = \max_{c,\,a',\,k',\,l'} \frac{c^{1-\theta}}{1-\theta} + \beta\,\mathbb{E}\,V(m',z')$$

Constraints and laws of motion

consumption–savings:	$c + a' = m$
capital:	$k' = a' + b', \qquad b' \le \xi\, k'$
cash-on-hand:	$m' = \phi W + (1+r)\,a' + \pi'$

$$\pi(\text{profit}) = \underbrace{z'\,\varepsilon'\left(k'^{\alpha}\, l'^{1-\alpha}\right)^{\eta}}_{\text{output } y',\ \eta<1} - \underbrace{Wl'}_{\text{wage bill}} - \underbrace{Rk'}_{\text{user cost, } R=r+\delta}$$

z', ϵ' are fat tailed drawn from a mixture of normals, and k', l' chosen *before* (z', ε') realize

Boar–Gorea–Midrigan: Implications

Spanish Orbis + EFF: estimate
...then compare to a frictionless planner$\Rightarrow$ aggregate losses relative to the efficient allocation.

Related: what explains unused credit buffers among SMEs?

Aydin–Kim (2026); Amberg–Jacobson–Quadrini–Rogantini Picco (2026)

- **AK:** Turkish credit-line experiment—sales respond mainly among firms far from debt limits; borrowing rises only moderately $\Rightarrow$ capacity as **buffer**
- **AJQR:** Swedish credit register—low utilization even when rates $\ll$ ROE;

Open question: which force dominates the buffer—DDW financing risk vs. KT consumption risk? More to be done...

So far ...

Dynamic investment–financing models, with **precautionary savings** as a central prediction—spare debt capacity to finance future investments

Extension 5: what else generates precautionary savings?

Risk averse agents save to smooth their consumption We've seen these in a **unified framework**.

Extension 6: the flip side of this question:

What else do leverage dynamics matter for in a corporate setting? (Recruiting and retaining workers)

- **Michaels–Page–Whited (2019)**: firm-level investment and labor bargaining
- **Bai–Wang (2026)**: emphasize the GE effect and the implication on mis-allocation

Michaels–Page–Whited (2019): debt and the wage bargain

Environment (DDW + labor): risk-neutral firm; defaultable debt; linear hiring cost

$$P(k,n,b,z) = \max_{k',n',b'} \ \hat{D} + q\,\mathbb{E}\big[P(k',n',b',z')\big]$$

$$\text{output} = z'(k')^{\alpha}(n')^{\beta}, \quad \text{default if } z' < \hat{z}(k',n',b')$$

$$\text{Hiring cost} = c_n\,(n'-n)\,\mathbf{1}_{\{n'-n>0\}}$$

Wage bargain: Stole–Zwiebel (pair-wise, over marginal surplus)

$$\underbrace{(1-\omega)\frac{W}{n'} + \omega\,\frac{\partial W}{\partial n'}}_{\text{marginal wage}} = \underbrace{\omega\,q\,\mathbb{E}\Big[\frac{\partial a'}{\partial n'}\,\Big|\,z' \geq \hat{z}\Big]}_{\text{worker's contribution (if survive)}} + \underbrace{(1-\omega)\,\ell}_{\text{outside option}}$$

Debt channel: $\uparrow b' \ \Rightarrow \uparrow \hat{z} \ \Rightarrow$ fewer survival states in $\mathbb{E}[\partial a'/\partial n'] \ \Rightarrow \downarrow$ bargained W.

So what shapes leverage? Beyond DDW, debt serves **other strategic roles** for firm.

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Thank you

Questions?